CLAIMS

What is claimed is:

1. A method for representing a fractional number in a digital system, comprising the steps of:

    - separating the fractional number into an integer part and a decimal part;
    - encoding the integer part using a parametric series representation;
    - representing the decimal part as a sum of one or more scaled members of a pre-determined convergent mathematical series; and
    - combining the encoded integer part and the representation of the decimal part into a composite data structure.

2. The method of claim 1, wherein the step of representing the decimal part further comprises:

    - for each scaled member of the sum, encoding it as a pair of indices into at least two pre-shared matrices, one containing multipliers and one containing members of the convergent series.

3. The method of claim 2, wherein the convergent mathematical series is the Basel problem series.

4. The method of claim 1, wherein the composite data structure is an N-gram of states in a high-cardinality, parallel data interface.

5. A system for representing a fractional number, comprising:

    - a module for separating a fractional number into an integer part and a decimal part;

- a first encoding module for encoding the integer part as a parametric series representation;
- a second encoding module for representing the decimal part as a sum of scaled members of a convergent series, wherein each scaled member is encoded as a pair of indices into pre-shared matrices; and
- a module for combining the outputs of the first and second encoding modules into a composite N-gram data structure for transmission or storage.

6. The system of claim 5, wherein the pre-shared matrices comprise a matrix of multipliers and a matrix of series members.